Chapter 11. Relational Inference Matched to the Data Structure

Graphical models, Lecture 4 (hidden confounders, FCI and PAGs, nonlinear CI tests, PCMCI+, feedback)

Contents

1 PC Assumes All Common Causes Are Observed **2**

1.1 Observational data may not distinguish three explanations 2

2 FCI Infers a PAG Instead of a DAG **2**

2.1 Ancestral graphs and PAGs 2

2.2 Reading PAG endpoints 3

2.3 FCI can still find the arrowheads of colliders 3

2.4 FCI checks wider separating sets, then orients conservatively 3

3 When Basic Assumptions Break, Methods Change **4**

3.1 Nonlinear relations: change the independence test 4

3.2 Time series: PCMCI+ for lagged relations 4

3.3 Feedback: the DAG assumption fails 4

4 Choosing a Method for the Data Structure **5**

5 Selected exercises **5**

1 PC Assumes All Common Causes Are Observed

PC premises: one DAG, no missing common cause, Markov and faithfulness, correct CI tests.

Hidden common cause: $E \leftarrow U \rightarrow P$ with U unobserved makes E, P related without a direct link, and PC may output a direct edge. **With possible hidden common causes, PC edges are not direct relations.**

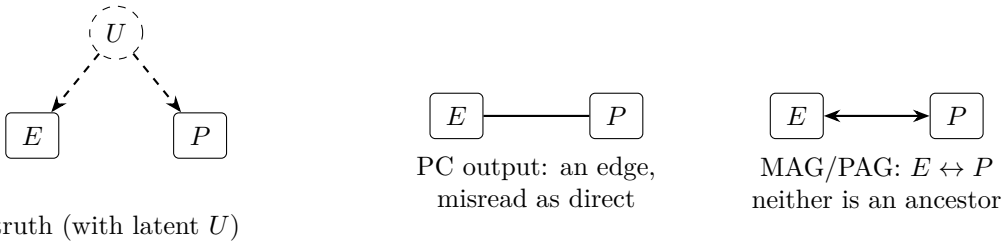


Figure 1: A hidden common cause: a DAG on the observed variables must pick a direction; FCI's ancestral graphs use a bidirected edge.

1.1 Observational data may not distinguish three explanations

Only X, Y observed and related:

Possible structure	Explanation	Observed conditional independences
$X \rightarrow Y$	X causes Y	none
$X \leftarrow Y$	Y causes X	none
$X \leftarrow U \rightarrow Y$	hidden common cause U	none

More observed variables can exclude some explanations.

2 FCI Infers a PAG Instead of a DAG

2.1 Ancestral graphs and PAGs

Marginalizing latents from a DAG gives a **maximal ancestral graph** (MAG): $X \rightarrow Y$ means X is an ancestor of Y; $X \leftrightarrow Y$ means neither is an ancestor of the other. Equivalent MAGs are summarized by a **partial ancestral graph** (PAG): marks common to all MAGs are shown, others are circles $\circ$.

	Object represented	Meaning of relations
PC's CPDAG	DAG equivalence class under the assumption of no missing common causes among analyzed variables	only directions common to all equivalent DAGs are fixed
FCI's PAG	equivalence class of observed-variable structures allowing hidden common causes	only endpoints common to all compatible structures are fixed

FCI uses CIs but does *not* assume causal sufficiency.

2.2 Reading PAG endpoints

PAG edge	How to read it
$X \to Y$	apart from selection bias, in every compatible structure X is an ancestor of Y and Y is not an ancestor of X
$X \leftrightarrow Y$	neither is an ancestor of the other; compatible with a structure with a hidden common cause
$X \circ\!\!\to Y$	it is certain that Y is not an ancestor of X; the mark at X is undetermined
$X \circ\!\!-\!\!\circ Y$	the two variables are adjacent but both endpoints are undetermined

Arrowhead = *not an ancestor*; circle = tail or arrowhead.

2.3 FCI can still find the arrowheads of colliders

Skeleton $X - Y - Z$, $X \not\sim Z$, $X \perp\!\!\!\perp Z$ but $X \not\perp\!\!\!\perp Z \mid Y$: FCI orients

$$X \circ\!\!\to Y \leftarrow\!\!\circ Z,$$

leaving open whether X causes Y or shares a hidden cause with it.

2.4 FCI checks wider separating sets, then orients conservatively

Algorithm: FCI

Edge removal.

1. Start from the complete graph on the observed variables.
2. If $X \perp\!\!\!\perp Y \mid S$ is found, remove $X - Y$ (as in PC's skeleton phase).
3. Widen the candidate separating sets beyond current neighbors to the variables on possible paths (Possible-D-Sep) and check again.

Endpoint determination.

1. Record the separating set used when an edge was removed.
2. Mark as colliders the middle variables not contained in the separating set.
3. Propagate only forced endpoints (Zhang's rules R1–R10).

Undetermined endpoints remain circles $\circ$.

Remark: why neighbors are not enough

In a MAG a separating set may need a node W reaching X only via colliders (e.g. $W \leftrightarrow V \leftrightarrow X$), not adjacent to X or Y. FCI uses the computable superset Possible-D-Sep; PC would wrongly keep the edge.

RFCI skips most of Possible-D-Sep with targeted triple tests: faster, slightly less informative, still sound.

3 When Basic Assumptions Break, Methods Change

Basic assumption	Situation where it fails	Candidate methods
all common causes observed	unobserved common causes exist	FCI, RFCI
the chosen test captures the relation	partial correlation misses nonlinear relations	KCI, distance correlation
independent observations	autocorrelation and time lags	PCMCI+
no cycles in the graph	feedback among variables	CCD, cyclic LiNGAM

3.1 Nonlinear relations: change the independence test

Example: dependence with zero correlation

$X \sim \mathrm{Unif}(-1, 1)$, $Y = X^2 + \varepsilon$, $\varepsilon \perp\!\!\!\perp X$: $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ by symmetry, so Fisher-z and PC remove a real edge.

CI test	When to use
Fisher-z	linear conditional relations in continuous Gaussian data
kernel CI test (KCI)	general nonlinear conditional relations (based on cross-covariance operators in reproducing-kernel spaces)
distance correlation	nonlinear relations without strong assumptions on the distribution; $\mathrm{dCor}(X, Y) = 0$ iff $X \perp\!\!\!\perp Y$

PC/FCI stay the same; only the test changes. Nonparametric tests lose power in small samples and are slower.

3.2 Time series: PCMCI+ for lagged relations

Target relations:

$$X^j_{t-\tau} \longrightarrow X^k_t, \qquad \tau = 0, 1, \ldots, \tau_{\max}.$$

Algorithm: PCMCI+ (outline)

1. Among the past values of all variables, first reduce the candidate parents that may be related to each present value (a PC-type condition-selection step on lagged links).
2. Re-test each candidate link conditioning on the other past parents of both endpoints (momentary conditional independence, MCI), which removes autocorrelation effects.
3. Distinguish time-lagged relations from same-time (contemporaneous) relations and represent them in a time-series graph; contemporaneous links are oriented with PC-type rules.

Example output: $X^1_{t-2} \to X^2_t$, $X^2_{t-1} \to X^3_t$. Ordinary PC on autocorrelated series gives false edges (Exercise 11.2); check $\tau_{\max}$ and stationarity.

3.3 Feedback: the DAG assumption fails

E.g. $X_1 \to X_2 \to X_3 \to X_1$.

Method	Core idea
CCD	uses conditional independences to find relations common to the equivalence class of cyclic graphs
cyclic LiNGAM	uses linearity and non-Gaussian errors to further identify the cyclic structure

$X = BX + \epsilon$ is an equilibrium; it needs $I - B$ invertible (stability: $\varrho(B) < 1$). First decide whether feedback is *simultaneous* or *time-lagged* (PCMCI+).

4 Choosing a Method for the Data Structure

Feature of the data	Method	Main points to check
hidden common causes suspected	FCI or RFCI	read PAG circles and arrowheads separately
nonlinear relations expected	PC/FCI with nonparametric tests	test power and computing time
time lags and autocorrelation	PCMCI+	maximal lag and stationarity
simultaneous feedback possible	CCD or cyclic LiNGAM	distinguish cycles from time lags

Summary

- With hidden confounders FCI outputs a PAG: arrowhead = not an ancestor, circle = undetermined.
- FCI searches Possible-D-Sep; RFCI trades information for speed.
- Nonlinear: KCI/dCor tests; time series: PCMCI+; feedback: CCD, cyclic LiNGAM.

5 Selected exercises

Motivation. A hidden cause mimicking a direct effect, spurious correlation of autocorrelated series, and the equilibrium of a cyclic SEM.

Exercise 11.1 (A hidden common cause mimics a direct effect)

Let $U \sim N(0,1)$ be latent, $X = aU + \epsilon_X$, $Y = bU + \epsilon_Y$, with $U, \epsilon_X, \epsilon_Y$ independent, $\epsilon_X \sim N(0, \sigma_X^2)$, $\epsilon_Y \sim N(0, \sigma_Y^2)$, $\sigma_Y^2 > 0$.
(a) Find c and $\tau^2 > 0$ such that the SEM $X' \sim N(0, a^2 + \sigma_X^2)$, $Y' = cX' + \eta$, $\eta \sim N(0, \tau^2)$ independent of X', has the same distribution as (X, Y).
(b) Add an observed $Z \sim N(0,1)$, independent of $(U, \epsilon_X, \epsilon_Y)$, with $X = dZ + aU + \epsilon_X$, $d \neq 0$. Show that $Z \perp\!\!\!\perp Y$, whereas in any Gaussian SEM $Z \to X \to Y$ with nonzero coefficients Z and Y are dependent.

Solution [Supplement].

(a)

Step 1. (X, Y) is centered Gaussian with $\mathrm{Var}X = a^2 + \sigma_X^2$, $\mathrm{Var}Y = b^2 + \sigma_Y^2$, $\mathrm{Cov}(X, Y) = ab$, and is determined by this covariance.

Step 2. $\mathrm{Cov}(X', Y') = c(a^2 + \sigma_X^2)$, $\mathrm{Var}Y' = c^2(a^2 + \sigma_X^2) + \tau^2$; matching,

$$c = \frac{ab}{a^2 + \sigma_X^2}, \qquad \tau^2 = b^2 + \sigma_Y^2 - \frac{a^2 b^2}{a^2 + \sigma_X^2} = \sigma_Y^2 + \frac{b^2 \sigma_X^2}{a^2 + \sigma_X^2} > 0.$$

So (X, Y) alone cannot tell $X \leftarrow U \rightarrow Y$ from $X \rightarrow Y$ (or $Y \rightarrow X$).

(b)

Step 3. $Y = bU + \epsilon_Y$ is a function of $(U, \epsilon_Y) \perp\!\!\!\perp Z$, so $Z \perp\!\!\!\perp Y$.

Step 4. If $X = dZ + \epsilon_X'$, $Y = cX + \eta$ with $c, d \neq 0$, then $\mathrm{Cov}(Z, Y) = cd \neq 0$.

This CI pattern is what FCI uses to orient $Z \circ\!\!\rightarrow X \leftarrow\!\!\circ Y$, ruling out $X \rightarrow Y$. ■

Exercise 11.2 (Independent but autocorrelated series)

Let $X_t = \phi X_{t-1} + u_t$ and $Y_t = \phi Y_{t-1} + v_t$ be stationary AR(1) processes, $|\phi| < 1$, where (u_t) and (v_t) are i.i.d. $(0, 1)$ sequences independent of each other. Let $C_n = \frac{1}{n} \sum_{t=1}^{n} X_t Y_t$. Show that $\mathbb{E}C_n = 0$ and

$$n\mathrm{Var}(C_n) \longrightarrow \gamma(0)^2 \frac{1 + \phi^2}{1 - \phi^2}, \qquad \gamma(0) = \frac{1}{1 - \phi^2},$$

whereas for $\phi = 0$ the limit is $\gamma(0)^2 = 1$.

Solution [Supplement].

Step 1. $X_t = \sum_{m \geq 0} \phi^m u_{t-m}$ has mean 0 and $\gamma(h) = \phi^{|h|}/(1 - \phi^2)$, likewise Y; by independence $\mathbb{E}(X_t Y_t) = 0$.

Step 2. By independence,

$$\mathrm{Cov}(X_s Y_s, X_t Y_t) = \mathbb{E}(X_s X_t)\, \mathbb{E}(Y_s Y_t) = \gamma(t - s)^2.$$

Step 3. Hence

$$n\mathrm{Var}(C_n) = \frac{1}{n} \sum_{s=1}^{n} \sum_{t=1}^{n} \gamma(t - s)^2 = \sum_{|h| < n} \Big(1 - \frac{|h|}{n}\Big) \gamma(h)^2,$$

and since $\sum_h \gamma(h)^2 < \infty$, dominated convergence gives

$$\sum_{h \in \mathbb{Z}} \gamma(h)^2 = \gamma(0)^2 \Big(1 + 2 \sum_{h \geq 1} \phi^{2h}\Big) = \gamma(0)^2 \Big(1 + \frac{2\phi^2}{1 - \phi^2}\Big) = \gamma(0)^2 \frac{1 + \phi^2}{1 - \phi^2}.$$

So $\mathrm{Var}(r_n) \approx \frac{1}{n} \cdot \frac{1 + \phi^2}{1 - \phi^2}$ (19× the i.i.d. value at $\phi = 0.9$): naive Fisher-z declares spurious edges. ■

Exercise 11.3 (Equilibrium of a linear cyclic SEM)

Let B be a $p \times p$ matrix with spectral radius $\varrho(B) < 1$.
(a) Show that $I - B$ is invertible and $(I - B)^{-1} = \sum_{m \geq 0} B^m$.
(b) Show that for a fixed ϵ and any start $X^{(0)}$, the iteration $X^{(t+1)} = BX^{(t)} + \epsilon$ converges to $(I - B)^{-1}\epsilon$.
(c) For the feedback loop $X_1 = bX_2 + \epsilon_1$, $X_2 = aX_1 + \epsilon_2$, show that $\varrho(B) < 1$ iff $|ab| < 1$, and find the equilibrium.

Solution [Supplement].

(a)

Step 1. Gelfand: $\varrho(B) = \lim_m \|B^m\|^{1/m}$, so for $\varrho(B) < q < 1$, $\|B^m\| \leq q^m$ for $m \geq m_0$; $\sum_m B^m$ converges.

Step 2. $(I - B)\sum_{m=0}^{M} B^m = I - B^{M+1} \to I$, so $(I - B)\sum_{m\geq 0} B^m = I$, and likewise in the other order.

(b) $X^{(t)} = B^t X^{(0)} + \sum_{m=0}^{t-1} B^m \epsilon \to (I - B)^{-1}\epsilon$ by Step 1.

(c) $B = \begin{pmatrix} 0 & b \\ a & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 - ab$, eigenvalue moduli $\sqrt{|ab|}$, so $\varrho(B) < 1 \iff |ab| < 1$, and

$$X_1 = \frac{\epsilon_1 + b\epsilon_2}{1 - ab}, \qquad X_2 = \frac{a\epsilon_1 + \epsilon_2}{1 - ab}.$$

■